

Piecewise Linear Manifolds

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Abstract

Contents

1	Triangulated Manifolds	3
1.1	Piecewise Linear Manifolds	3
1.2	Combinatorial Manifolds	3
2	Pseudomanifolds	5
2.1	Basic Properties	5
3	Intersection Homology	6
3.1	Perversities	6
3.2	Simplicial Intersection Homology	6

1 Manifolds with Combinatorial Structures

1.1 Triangulated Manifolds

Definition 1.1.1 (Triangulated Manifolds)

Let M be a topological manifold. We say that M is a triangulated manifold if M is triangulated as a topological space.

Theorem 1.1.2 (Edwards)

Let $n \geq 5$. Let $\mathcal{K}$ be a simplicial complex of dimension n . Then $|\mathcal{K}|$ is a topological manifold of dimension n if and only if the following are true.

- $H_*(\text{lk}(\sigma)) \cong H_*(S^{n-|\sigma|})$ for all $\emptyset \neq \sigma \in \mathcal{K}$.
- $\text{lk}(v)$ is simply connected for all $v \in V(\mathcal{K})$.

manifold Hauptvermutung: if $\mathcal{K}$ and $\mathcal{L}$ are simplicial complexes such that $|\mathcal{K}|$ and $|\mathcal{L}|$ are homeomorphic triangulated manifold, then is $\mathcal{K}$ combinatorially equivalent to $\mathcal{K}$? This is proven to be false for dimension ≥ 5 .

1.2 Piecewise Linear Manifolds

Proposition 1.2.1

The set

$$\Gamma = \{f : U \rightarrow V \mid U, V \subseteq \mathbb{R}^n \text{ are open, } f \text{ is a PL homeomorphism}\}$$

defines a pseudogroup of transformations on $\mathbb{R}^n$.

Definition 1.2.2 (Piecewise Linear Manifolds)

A piecewise linear manifold is a manifold whose pseudogroup of transformations is given by Γ .

1.3 Combinatorial Manifolds

Definition 1.3.1 (Combinatorial Manifolds)

Let M be a topological manifold. We say that M is a combinatorial manifold if there exists a simplicial complex $\mathcal{K}$ and a homeomorphism $|\mathcal{K}| \cong X$ such that

$$\text{lk}(\sigma) \cong_{\text{PL}} \partial \Delta^{\dim(M)-|\sigma|+1} \quad \text{or} \quad \Delta^{\dim(M)-|\sigma|}$$

for all $\sigma \in \mathcal{K}$.

Lemma 1.3.2

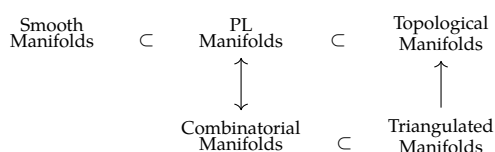
Let $\mathcal{K}$ be a simplicial complex. Suppose that there exists $n \in \mathbb{N}^+$ such that

$$\text{lk}(\sigma) \cong_{\text{PL}} \partial \Delta^{n-|\sigma|+1} \quad \text{or} \quad \Delta^{n-|\sigma|}$$

for all $\sigma \in \mathcal{K}$. Then $|\mathcal{K}|$ is a piecewise linear manifold of dimension n .

Theorem 1.3.3 (Whitehead 1940)

Let M be a piecewise linear manifold. Then M is a combinatorial manifold.



Note: in toric topology, PL manifold means combinatorial manifolds.

Theorem 1.3.4 (PL Hauptvermutung for combinatorial equivalence)

Let $\mathcal{K}$ and $\mathcal{L}$ be simplicial complexes such that $|\mathcal{K}|$ and $|\mathcal{L}|$ are combinatorial manifolds. Then $|\mathcal{K}| \cong_{\text{PL}} |\mathcal{L}|$ if and only if $\mathcal{K}$ and $\mathcal{L}$ are combinatorially equivalent.

2 Pseudomanifolds

2.1 Basic Properties

Definition 2.1.1 (Pseudomanifolds From Simplicial Complexes)

Let $\mathcal{K}$ be a simplicial complex. We say that $\mathcal{K}$ is a pseudomanifold of dimension n if the following are true.

- All maximal simplices of $\mathcal{K}$ are of dimension n .
- Each $(n-1)$ -simplex of $\mathcal{K}$ is the face of two n -simplices of $\mathcal{K}$.
- For any two n -simplices $\sigma, \sigma' \in \mathcal{K}$, there exists n -simplices $\sigma = \sigma_0, \dots, \sigma_k = \sigma'$ such that $\sigma_i \cap \sigma_{i+1}$ is an $(n-1)$ -simplex for $0 \leq i \leq k-1$.

Definition 2.1.2 (Pseudomanifolds in General)

Let X be a space. We say that X is a pseudo-manifold if X is triangulated by a simplicial complex $\mathcal{K}$ that is a pseudo-manifold.

Proposition 2.1.3

Let M be a combinatorial manifold. Then M is a pseudomanifold.

There are inclusions

$$\text{PL Manifolds} \subset \text{Combinatorial Manifold} \subset \text{Pseudomanifolds}$$

Lemma 2.1.4

Let X be a pseudomanifold of dimension n . Then $H_n(X) \cong \mathbb{Z}$ or 0 .

Lemma 2.1.5

Let $\mathcal{K}$ be a triangulation of S^n . Then the following are true.

- $\mathcal{K}$ is a pseudomanifold.
- If $\mathcal{K}$ is finite, then the generator of $H_n(\mathcal{K})$ is represented by the sum of n -simplices of $\mathcal{K}$.

3 Intersection Homology

3.1 Perversities

Definition 3.1.1 (Perversities)

A perversity is a function $p : \{2, \dots, m\} \rightarrow \mathbb{N}$ such that the following are true.

- $p(2) = 0$.
- $p(i+1) = p(i)$ or $p(i) + 1$.

Definition 3.1.2 (Special Perversities)

Define the following perversities.

- The zero perversity is given by $i \mapsto 0$ for all i .
- The top perversity $t : \{2, \dots, m\} \rightarrow \mathbb{N}$ is given by $t(i) = i - 2$.
- The lower middle perversity $m : \{2, \dots, m\} \rightarrow \mathbb{N}$ is given by $m(i) = \lfloor i/2 \rfloor - 1$.
- The upper middle perversity $n : \{2, \dots, m\} \rightarrow \mathbb{N}$ is given by $n(i) = \lceil i/2 \rceil - 1$.

3.2 Simplicial Intersection Homology

4 Intersection Homology Theory

4.1 Stratified Spaces

(cant be just a filtration)

4.2

Definition 4.2.1 (Singular p-Allowable Simplices)

Let X be a stratified space with filtration $(X_k)_{0 \leq k \leq m}$. Let $p : \{2, \dots, m\} \rightarrow \mathbb{N}$ be a perversity. Define the chain complex of singular p -allowable simplices to be the subcomplex given by

$$I^p C_i(X) = \mathbb{Z} \left\langle (\sigma : \Delta^i \rightarrow X) \in C_i(X) \mid \sigma^{-1}(X_{m-j} \setminus X_{m-j-1}) \subseteq \text{sk}_{i-j+p(j)} \Delta^i \text{ for all } j \right\rangle \leq C_i(X)$$

Definition 4.2.2 (Intersection Homology)

Let X be a stratified space. Let $p : \{2, \dots, m\} \rightarrow \mathbb{N}$ be a perversity. Define the i th intersection homology group of X with perversity p to be

$$I^p H_i(X) = H_i(I^p C_\bullet(X))$$

Definition 4.2.3 (Intersection Cohomology)

Let X be a stratified space. Let $p : \{2, \dots, m\} \rightarrow \mathbb{N}$ be a perversity. Define the i th intersection cohomology group of X with perversity p to be

$$I^p H^i(X) = H_i(\text{Hom}_{\mathbb{Z}}(I^p C_\bullet(X), \mathbb{Z}))$$

Lemma 4.2.4

Let X be a stratified space. Let $p : \{2, \dots, m\} \rightarrow \mathbb{N}$ be a perversity. Then there are isomorphisms

$$I^p H_i(X) \cong H_i(X)$$

Relative intersection homology.

Excision.

Mayer-vietoris sequence.

Homotopy invariance.

compact support

4.3 Poincare Duality